

# Lab Report 9 – Collisions in 1D

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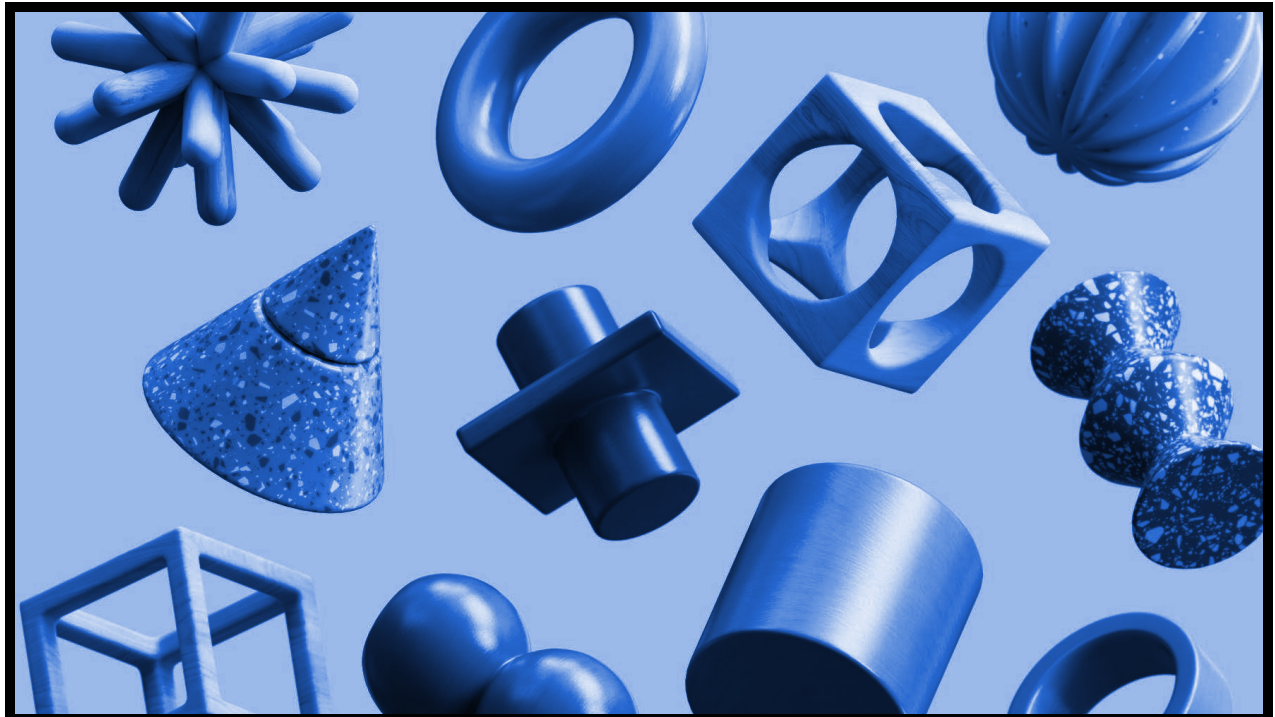
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## Objectives

To confirm the conservation of total momentum in all collisions, and to confirm the expected changes in kinetic energy for different types of collisions.

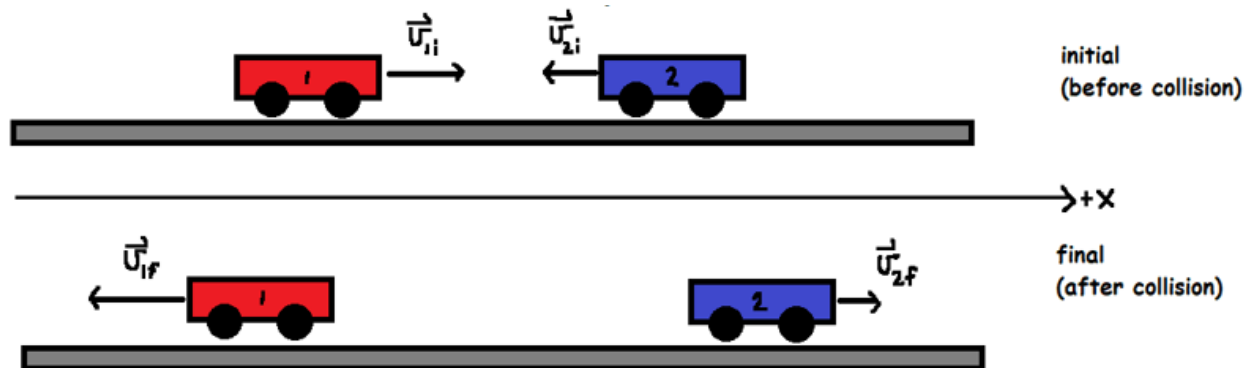
## Equipments

The list of equipments used in this experiment is:

- Carts
- Rail
- Motion Sensors
- Computer Interface
- Triple Beam Balance
- Cart Masses

## Theory and Equations

Consider a 1-D collision between 2 objects:



### 1. Total Initial Momentum

$$P_{tot,i,x} = m_1 v_{1ix} + m_2 v_{2ix} \quad (\text{Eq. 1})$$

### 2. Total Final Momentum

$$P_{tot,f,x} = m_1 v_{1fx} + m_2 v_{2fx} \quad (\text{Eq. 2})$$

**3. Conserved Total Momentum**

$$P_{tot,i,x} = P_{tot,f,x} \quad (\text{Eq. 3})$$

**4. Percentage Difference**

$$\text{Percent Difference} = \left| \frac{P_{tot,f,x} - P_{tot,i,x}}{\frac{1}{2}(P_{tot,f,x} + P_{tot,i,x})} \right| * 100\% \quad (\text{Eq. 4})$$

**5. Total Initial Kinetic Energy**

$$K_{tot,i} = \frac{1}{2}m_1v_{1ix}^2 + \frac{1}{2}m_2v_{2ix}^2 \quad (\text{Eq. 5})$$

**6. Total Final Kinetic Energy**

$$K_{tot,f} = \frac{1}{2}m_1v_{1fx}^2 + \frac{1}{2}m_2v_{2fx}^2 \quad (\text{Eq. 6})$$

**7. Change in Kinetic Energy**

$$\Delta K = K_{tot,f} - K_{tot,i} \quad (\text{Eq. 7})$$

$\Delta K > 0$  : Explosive

$\Delta K \sim 0$  : Elastic

$\Delta K < 0$  : Inelastic

**8. Percentage Change in Kinetic Energy**

$$\text{Percent Change} = \left( \frac{K_{tot,f} - K_{tot,i}}{K_{tot,i}} \right) * 100\% \quad (\text{Eq. 8})$$

**Summary of Procedures**

First, we set up a stable surface using the above equipment as shown in the illustration and measure the mass of each cart in the following specific situations: blue cart without added mass and red cart with/without added mass. Next, We need to set up the motion sensors with the computer interface so that both are aligned with the same x-axis to form a 1-D dimension. We

move a random cart on the rail several times to check whether the computer interface works properly. For the experiment, we will address the following scenarios:

1. Elastic collision (no added masses on both carts)
2. Elastic collision with the red cart having added mass.
3. Fully inelastic collision (no added masses on both carts)
4. Fully inelastic collision with the red cart having added mass
5. Explosive collision with the red cart having added mass and the blue cart having armed and released spring-loaded plunger.

For each of the above scenarios, we use a computer interface to measure the initial and final velocity of each cart.

## Data and Observations

| Mass (g) |                      |                              |
|----------|----------------------|------------------------------|
| Blue car | Red car + added Mass | Red car (without added Mass) |
| 253.6    | 512.2                | 255.0                        |

| Case                                             | Red Car                |                                | Blue Car               |                                |
|--------------------------------------------------|------------------------|--------------------------------|------------------------|--------------------------------|
|                                                  | Initial Velocity (m/s) | After-collision Velocity (m/s) | Initial Velocity (m/s) | After-collision Velocity (m/s) |
| Elastic Collision (no added Mass)                | 0.42                   | -0.03                          | 0                      | 0.42                           |
| Elastic Collision (1 car has added Mass)         | 0                      | 0.34                           | 0.58                   | -0.22                          |
| Fully Inelastic Collision (no added Mass)        | 0                      | 0.49                           | 1.02                   | 0.50                           |
| Fully Inelastic Collision (1 car has added Mass) | 0                      | 0.25                           | 0.84                   | 0.26                           |
| Explosive Collision (with added Mass)            | 0                      | 0.64                           | 0.69                   | -0.51                          |

## Data Analysis

### Sample Calculation:

| Mass (g) |                      |                              |
|----------|----------------------|------------------------------|
| Blue car | Red car + added Mass | Red car (without added Mass) |
| 253.6    | 512.2                | 255.0                        |

| Case                              | Red Car                |                                | Blue Car               |                                |
|-----------------------------------|------------------------|--------------------------------|------------------------|--------------------------------|
|                                   | Initial Velocity (m/s) | After-collision Velocity (m/s) | Initial Velocity (m/s) | After-collision Velocity (m/s) |
| Elastic Collision (no added Mass) | 0.42                   | -0.03                          | 0                      | 0.42                           |

$$\begin{aligned} \text{Total Initial Momentum} = P_{tot,i,x} &= m_1 v_{1ix} + m_2 v_{2ix} = (253.6\text{g})(0\text{m/s}) + (255.0\text{g})(0.42\text{m/s}) \\ &= 107 = 1.1\text{E}+2 \text{ (g.m/s)} \end{aligned}$$

$$\begin{aligned} \text{Total Final Momentum} = P_{tot,f,x} &= m_1 v_{1fx} + m_2 v_{2fx} = (253.6\text{g})(0.42\text{m/s}) + (255.0\text{g})(-0.03\text{m/s}) \\ &= 98.862 = 9.9\text{E}+1 \text{ (g.m/s)} \end{aligned}$$

$$\text{Percent Difference} = \left| \frac{P_{tot,f,x} - P_{tot,i,x}}{\frac{1}{2}(P_{tot,f,x} + P_{tot,i,x})} \right| * 100\% = \left| \frac{1.1\text{E}2 \text{ g.m/s} - 9.9\text{E}1 \text{ g.m/s}}{\frac{1}{2}(1.1\text{E}2 \text{ g.m/s} + 9.9\text{E}1 \text{ g.m/s})} \right| * 100\% = 8.0 \%$$

$$\begin{aligned} \text{Total Initial Kinetic Momentum} = K_{tot,i} &= \frac{1}{2}m_1 v_{1ix}^2 + \frac{1}{2}m_2 v_{2ix}^2 = \\ &= \frac{1}{2} (253.6\text{g})(0\text{m/s})^2 + \frac{1}{2} (255.0\text{g})(0.42\text{m/s})^2 = 22.491 = 2.2\text{E}+1 \text{ mJ} \end{aligned}$$

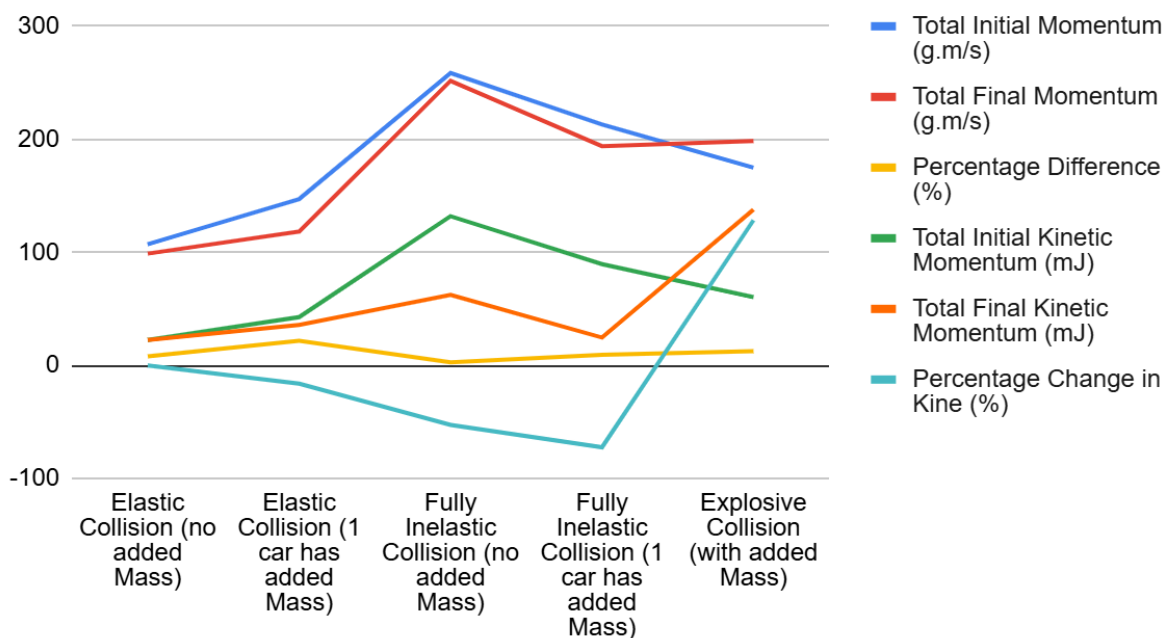
$$\begin{aligned} \text{Total Final Kinetic Momentum} = K_{tot,f} &= \frac{1}{2}m_1 v_{1fx}^2 + \frac{1}{2}m_2 v_{2fx}^2 = \\ &= \frac{1}{2} (253.6\text{g})(0.42\text{m/s})^2 + \frac{1}{2} (255.0\text{g})(-0.03\text{m/s})^2 = 22.253 = 2.2\text{E}1 \text{ mJ} \end{aligned}$$

$$\begin{aligned} \text{Percentage Change in Kinetic Energy} &= \left( \frac{K_{tot,f} - K_{tot,i}}{K_{tot,i}} \right) * 100\% = \\ &= \left( \frac{2.2\text{E}1 \text{ mJ} - 2.2\text{E}+1 \text{ mJ}}{2.2\text{E}1 \text{ mJ}} \right) * 100\% = 0 \% \end{aligned}$$

| Total Initial Momentum (g.m/s) | Total Final Momentum (g.m/s) | Percentage Difference (%) | Total Initial Kinetic Momentum (mJ) | Total Final Kinetic Momentum (mJ) | Percentage Change in Kine (%) |
|--------------------------------|------------------------------|---------------------------|-------------------------------------|-----------------------------------|-------------------------------|
| 1.1E+2                         | 9.9E+1                       | 8.0                       | 2.2E+1                              | 2.2E+1                            | 0                             |
| 1.5E+2                         | 1.2E+2                       | 22                        | 4.3E+1                              | 3.6E+1                            | -16                           |
| 2.6E+2                         | 2.5E+2                       | 2.7                       | 1.3E+2                              | 6.2E+1                            | -53                           |
| 2.1E+2                         | 1.9E+2                       | 9.4                       | 8.9E+1                              | 2.5E+1                            | -73                           |
| 1.8E+2                         | 2.0E+2                       | 13                        | 6.0E+1                              | 1.4E+2                            | 1.3E+2                        |

## Results

Distribution of Momentum in Different Experimental Cases



## Discussion / Conclusion

When we look at the observational data table, we notice the final velocity of both carts in the inelastic collision cases. Regardless of whether or not we add mass to the red cart, one key

feature of the inelastic collision remains the same: after the collision, both carts move at the nearly same velocity. This happens because the 2 carts are physically attached to each other during the collision due to velcro tape, which prevents them from bouncing apart.

Furthermore, by analyzing the graph titled “Distribution of Momentum in Different Experimental Cases,” we observe a significant pattern that reinforces the main principles of momentum conservation. The blue line in the graph represents the “Total Initial Momentum,” while the red line represents the “Total Final Momentum.” Notably, these 2 lines are almost aligned with each other, indicating a small percentage difference, below 13% across all experimented cases. This minimal difference strongly supports the principle of momentum conservation, which states that the total momentum of a system remains constant before and after a collision. A clear demonstration of this principle can be seen in the first case of an elastic collision (without added mass). Since the 2 carts have nearly identical masses in this case, the interaction follows the expected momentum transfer dynamics. On collision, the red cart, which was initially moving, transfers nearly all of its momentum to the blue cart, which was previously at rest. As a result, the final velocity of the blue cart is nearly equal to the initial velocity of the red cart before the collision. At the same time, the red cart comes to a near-complete stop (at  $-0.03 \text{ m/s}$ ). This result is entirely consistent with the theoretical expectation of a perfectly elastic collision, in which momentum is completely transferred between objects of equal mass.

Finally, when we look at the Percentage Change in Kinetic Energy, we see that the results are fit to the theoretical expectation. Specifically, in the case of an elastic collision with no added mass, the percent change in kinetic energy is recorded as 0%, indicating a completely elastic interaction. This is consistent with the theoretical definition of an elastic collision, in which no kinetic energy is lost ( $\Delta K \sim 0$ ). Similarly, in the case of an elastic collision with added mass, the percent change is slightly negative, approximately -16%. Although not quite zero, this small deviation is still consistent with the theoretical principle that kinetic energy is still nearly conserved in an elastic collision. For inelastic collisions, we observe a negative percentage change in kinetic energy (at -53% and -73%), consistent with the theoretical framework stating that  $\Delta K < 0$  states inelastic collision. In these cases, some of the kinetic energy of the system might be converted into other energy forms. Finally, in the case of explosive collisions, the percentage change in kinetic energy is highly positive (at 128%). This is consistent with the

theoretical expectation that  $\Delta K > 0$  for explosive interactions, in which additional energy is introduced into the system, increasing the total kinetic energy after the collision.

## Post-Lab Question

At the beginning, 2 carts is at rest, so:

$$\text{Total Initial Momentum} = P_{tot,i,x} = 0$$

$v_{1fx}$  and  $v_{2fx}$  are corresponding to the after-collision velocity of cart 1 and cart 2, after the spring plunger is released. We have:

$$\text{Total Final Momentum} = P_{tot,f,x} = m_1 v_{1fx} + m_2 v_{2fx}$$

Given  $m_1 = 400g = 0.4kg$  and  $m_2 = 800g = 0.8kg$ , then:

$$P_{tot,f,x} = 0.4v_{1fx} + 0.8v_{2fx}$$

Since this situation assumes that total momentum is conserved, we have:

$$\text{Total Initial Momentum} = \text{Total Initial Momentum}$$

$$P_{tot,i,x} = P_{tot,f,x}$$

$$0 = m_1 v_{1fx} + m_2 v_{2fx}$$

$$0 = 0.4v_{1fx} + 0.8v_{2fx} \quad (1)$$

As all the potential energy (10.8 J) of the spring plunger is converted to the kinetic energy after the collision, we have:

$$K_{tot,f} = \frac{1}{2}m_1 v_{1fx}^2 + \frac{1}{2}m_2 v_{2fx}^2$$

$$10.8 = \frac{1}{2}0.4v_{1fx}^2 + \frac{1}{2}0.8v_{2fx}^2 \quad (2)$$

From (1), we have:

$$0 = 0.4v_{1fx} + 0.8v_{2fx} \Rightarrow v_{1fx} = -2v_{2fx} \quad (3)$$

Replace (3) into (2), we get:

$$10.8 = \frac{1}{2}0.4(4v_{2fx}^2) + \frac{1}{2}0.8v_{2fx}^2 \quad (4)$$

Solve the equations (4), we get:



$$v_{2fx} = 3 \text{ m/s}$$

$$v_{1fx} = 6 \text{ m/s}$$

In summary, the maximum speed of car 1 after releasing the spring plunger is 6 m/s. While car 2 gets a maximum speed of 3 m/s after releasing the spring plunger.